

# A Monero data model and API specification

*with implementations in Java, TypeScript, and C++  
using monero v0.18.5.3 'Flourine Fermi'*

Last updated October 04, 2026



Java implementation: <https://github.com/woodser/monero-java>

TypeScript implementation: <https://github.com/woodser/monero-ts>

C++ implementation: <https://github.com/woodser/monero-cpp>

Document source: <https://github.com/woodser/monero-java/blob/master/docs/monero-spec.drawio>

Document created and revisable using: <https://www.draw.io/>

## Monero Daemon Interface 1/2

setProxyUri(string proxyUri): void

isConnected(): bool

getVersion(): MoneroVersion

isTrusted(): bool

getHeight(): ulong

getBlockHash(ulong height): string

getBlockTemplate(string walletAddress, uint reserveSize=null): MoneroBlockTemplate

getLastBlockHeader(): MoneroBlockHeader

getBlockHeaderByHash(string blockHash): MoneroBlockHeader

getBlockHeaderByHeight(ulong height): MoneroBlockHeader

getBlockHeadersByRange(ulong startHeight=0, ulong endHeight=chainHeight): MoneroBlockHeader[]

getBlockByHash(string blockHash): MoneroBlock

getBlocksByHash(string[] blockHashes, ulong startHeight, bool prune=false): MoneroBlock[]

getBlockByHeight(ulong height): MoneroBlock

getBlocksByHeight(ulong[] heights): MoneroBlock[]

getBlocksByRange(ulong startHeight=0, ulong endHeight=chainHeight): MoneroBlock[]

getBlocksByRangeChunked(ulong startHeight=0, ulong endHeight=chainHeight, ulong maxChunkSize=3000000): MoneroBlock[]

getBlockHashes(string[] blockHashes, ulong startHeight): string[]

getTx(string txHash, bool prune=false): MoneroTx

getTxes(string[] txHashes, bool prune=false): MoneroTx[]

getTxHex(string txHash, bool prune=false): string

getTxHexes(string[] txHashes, bool prune=false): string[]

getMinerTxSum(ulong height, ulong numBlocks=chainHeight): MoneroMinerTxSum

getFeeEstimate(ulong graceBlocks=?): MoneroFeeEstimate

submitTxHex(string txHex, bool doNotRelay=false): MoneroSubmitTxResult

relayTxByHash(string txHash): void

relayTxesByHash(string[] txHashes): void

getTxPool(): MoneroTx[]

getTxPoolHashes(): string[]

getTxPoolBacklog(): MoneroTxBacklogEntry[]

getTxPoolStats(): MoneroTxPoolStats

flushTxPool(string[] txHashes=null): void

getKeyImageSpentStatus(string keyImage): MoneroKeyImageSpentStatus

getKeyImageSpentStatuses(string[] keyImages): MoneroKeyImageSpentStatus[]

getOutputs(MoneroOutput[] outputs): MoneroOutput[]

getInfo(): MoneroDaemonInfo

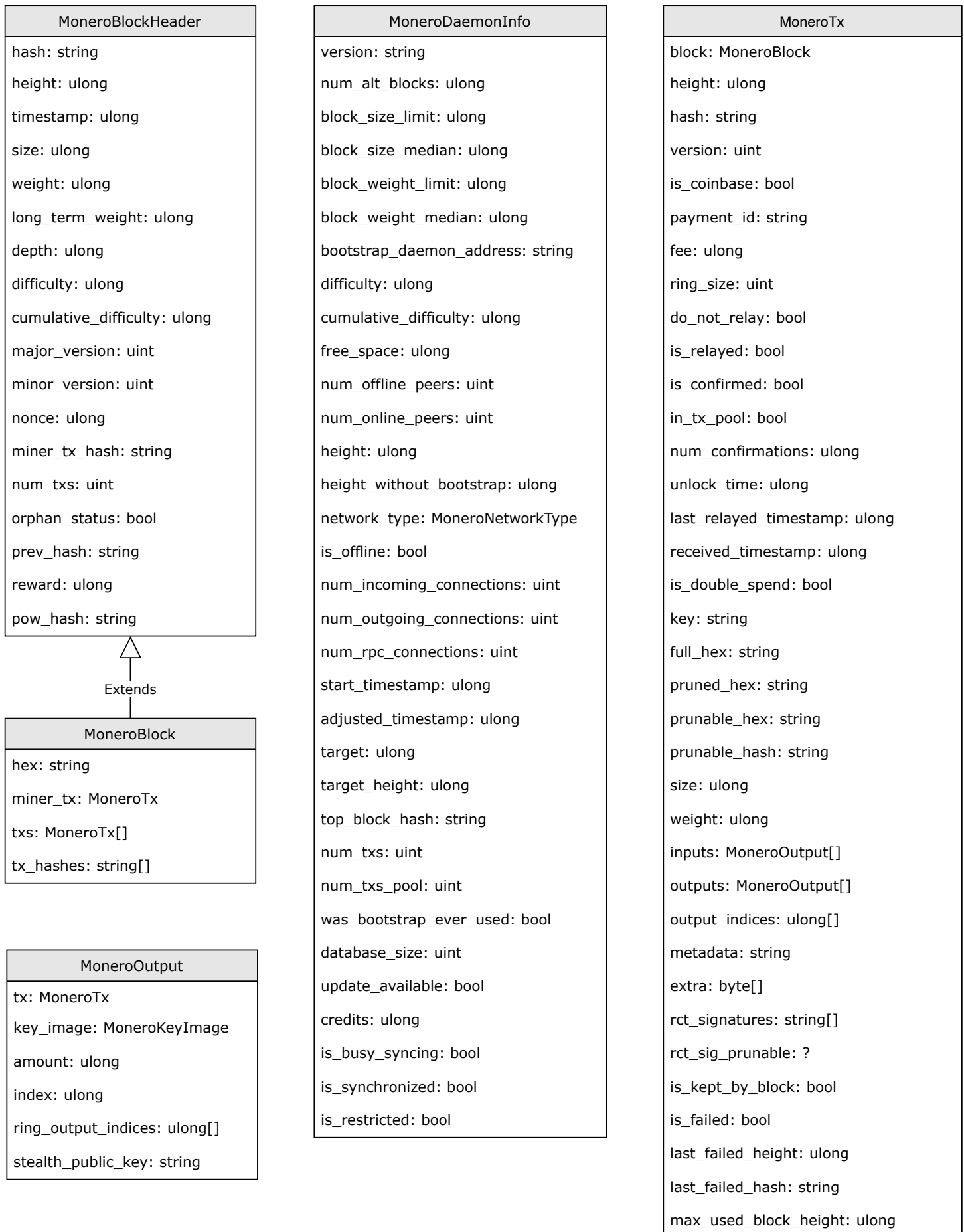
getSyncInfo(): MoneroDaemonSyncInfo

getHardForkInfo(): MoneroHardForkInfo

## Monero Daemon Interface 2/2

getAltChains(): MoneroAltChain[]  
getAltBlockHashes(): string[]  
getDownloadLimit(): uint  
setDownloadLimit(uint limit): uint  
resetDownloadLimit(): uint  
getUploadLimit(): uint  
setUploadLimit(uint): uint  
resetUploadLimit(): uint  
getPeers(): MoneroPeer[]  
getKnownPeers(): MoneroPeer[]  
setOutgoingPeerLimit(uint limit): void  
setIncomingPeerLimit(uint limit): void  
getPeerBans(): MoneroBan[]  
setPeerBan(MoneroBan ban): void  
setPeerBans(MoneroBan[] bans): void  
getOutputDistribution(ulong[] amounts, bool isCumulative, ulong startHeight=0, ulong endHeight=chainHeight):  
    MoneroOutputDistributionEntry[]  
getOutputHistogram(ulong[] amounts, ulong minCount, ulong maxCount, bool isUnlocked, ulong recentCutoff):  
    MoneroOutputHistogramEntry[]  
startMining(string address, ulong numThreads=null, bool isBackground=false, bool ignoreBattery=false): void  
stopMining(): void  
getMiningStatus(): MoneroMiningStatus  
submitBlock(string blockBlob): void  
submitBlocks(string[] blockBlobs): void  
checkForUpdate(): MoneroDaemonUpdateCheckResult  
downloadUpdate(string path): MoneroDaemonUpdateDownloadResult  
addListener(MoneroDaemonListener listener): void  
removeListener(MoneroDaemonListener listener): void  
waitForNextBlockHeader(): MoneroBlockHeader  
stop(): void  
pruneBlockchain(bool check): MoneroPruneResult

## Monero Daemon Types 1/3



|                             |
|-----------------------------|
| max_used_block_hash: string |
| signatures: string[]        |

## Monero Daemon Types 2/3

### MoneroDaemonSyncInfo

height: ulong  
connections: MoneroDaemonConnection[]  
spans: MoneroDaemonConnectionSpan[]  
target\_height: ulong  
next\_needed\_pruning\_seed: uint  
overview: ?

### MoneroBan

host: string  
ip: string  
is\_banned: bool  
seconds: ulong

### MoneroBlockTemplate

block\_template\_blob: string  
block\_hashing\_blob: string  
difficulty: ulong  
expected\_reward: ulong  
height: ulong  
prev\_hash: string  
reserved\_offset: ulong  
seed\_height: ulong  
seed\_hash: string  
next\_seed\_hash: string

### MoneroDaemonConnectionSpan

connection\_id: string  
num\_blocks: ulong  
remote\_address: string  
rate: ulong  
speed: ulong  
size: ulong  
start\_block\_height: ulong

### MoneroDaemonListener

onBlockHeader(MoneroBlockHeader header): void

### <enumeration> MoneroNetworkType

mainnet: 0  
testnet: 1  
stagenet: 2

### MoneroKeyImage

hex: string  
signature: string

### <enumeration> MoneroKeyImageSpentStatus

not\_spent: 0  
confirmed: 1  
tx\_pool: 2

### MoneroSubmitTxResult

is\_good: bool  
is\_relayed: bool  
is\_double\_spend\_seen: bool  
is\_fee\_too\_low: bool  
is\_mixin\_too\_low: bool  
has\_invalid\_input: bool  
has\_invalid\_output: bool  
has\_too\_few\_outputs: bool  
is\_rct: bool  
is\_overspend: bool  
is\_too\_big: bool  
sanity\_check\_failed: bool  
reason: string  
credits: ulong  
top\_block\_hash: string  
is\_tx\_result\_too\_big: bool  
nonzero\_unlock\_time: bool

### MoneroPeer

id: string  
address: string  
host: string  
port: uint  
is\_online: boolean  
last\_seen\_timestamp: ulong  
pruning\_seed: uint  
rpc\_port: uint  
rpc\_credits\_per\_hash: ulong  
type: ConnectionType  
num\_support\_flags: uint  
state: string  
send\_idle\_time: ulong  
receive\_idle\_time: ulong  
num\_sends: uint  
num\_receives: uint  
is\_local\_host: bool  
is\_local\_ip: bool  
live\_time: ulong  
is\_incoming: bool  
height: ulong  
current\_upload: ulong  
current\_download: ulong  
avg\_upload: ulong  
avg\_download: ulong

## Monero Daemon Types 3/3

| MoneroHardForkInfo     |
|------------------------|
| earliest_height: ulong |
| is_enabled: bool       |
| state: uint            |
| threshold: uint        |
| version: string        |
| num_votes: uint        |
| window: uint           |
| voting: uint           |
| credits: ulong         |
| top_block_hash: string |

| MoneroVersion    |
|------------------|
| number: uint     |
| is_release: bool |

| <enumeration><br>ConnectionType |
|---------------------------------|
| invalid: 0                      |
| ipv4: 1                         |
| ipv6: 2                         |
| tor: 3                          |
| i2p: 4                          |

| MoneroFeeEstimate       |
|-------------------------|
| fee: ulong              |
| fees: ulong[]           |
| quantizationMask: ulong |

| MoneroDaemonUpdateCheckResult |
|-------------------------------|
| is_update_available: bool     |
| version: string               |
| hash: string                  |
| auto_uri: string              |
| user_uri: string              |



| MoneroDaemonUpdateDownloadResult |
|----------------------------------|
| download_path: string            |

| MoneroMinerTxSum    |
|---------------------|
| emission_sum: ulong |
| fee_sum: ulong      |

| MoneroMiningStatus  |
|---------------------|
| is_active: bool     |
| is_background: bool |
| address: string     |
| speed: ulong        |
| num_threads: uint   |

| MoneroPruneResult  |
|--------------------|
| is_pruned: bool    |
| pruning_seed: uint |

| MoneroOutputHistogramEntry    |
|-------------------------------|
| amount: ulong                 |
| num_instances: ulong          |
| num_unlocked_instances: ulong |
| num_recent_instances: ulong   |

| MoneroTxPoolStats       |
|-------------------------|
| num_txs: uint           |
| num_not_relayed: uint   |
| num_failing: uint       |
| num_double_spends: uint |
| num_10m: uint           |
| fee_total: ulong        |
| bytes_max: ulong        |
| bytes_med: ulong        |
| bytes_min: ulong        |
| bytes_total: ulong      |
| histo: ?                |
| histo_98pc: ulong       |
| oldest_timestamp: ulong |

| MoneroAltChain                       |
|--------------------------------------|
| block_hashes: string[]               |
| difficulty: ulong                    |
| height: ulong                        |
| length: ulong                        |
| main_chain_parent_block_hash: string |

## Monero Wallet Interface 1/3

addListener(MoneroWalletListener listener): void  
removeListener(MoneroWalletListener): void  
getListeners(): MoneroWalletListener[]  
isViewOnly(): bool  
setDaemonConnection(MoneroDaemonConnection connection): void  
getDaemonConnection(): MoneroDaemonConnection  
isConnectedToDaemon(): bool  
getVersion(): MoneroVersion  
getNetworkType(): MoneroNetworkType  
getPath(): string  
getSeed(): string  
getSeedLanguage(): string  
getSeedLanguages(): string[]  
getPrivateSpendKey(): string  
getPrivateViewKey(): string  
getPublicSpendKey(): string  
getPublicViewKey(): string  
getPrimaryAddress(): string  
getAddress(uint accountId, uint subaddressIdx): string  
getAddressIndex(string address): MoneroSubaddress  
getDaemonConnection(): MoneroRpcConnection  
getDaemonHeight(): ulong  
getDaemonMaxPeerHeight(): ulong  
getApproximateChainHeight(): ulong  
getHeight(): ulong  
getHeightByDate(uint year, uint month, uint day): ulong  
sync(ulong startHeight=null, MoneroSyncListener listener=null): MoneroSyncResult  
startSyncing(): void  
stopSyncing(): void  
scanTxs(string[] txHashes): void  
rescanSpent(): void  
rescanBlockchain(): void  
getBalance(uint accountId=null, uint subaddressIdx=null): ulong  
getUnlockedBalance(uint accountId=null, uint subaddressIdx=null): ulong  
getAccounts(bool includeSubaddresses=false, string tag=null): MoneroAccount[]  
getAccount(uint accountId, bool includeSubaddresses=false): MoneroAccount  
createAccount(string label=null): void  
getSubaddresses(uint accountId, uint[] subaddressIndices): void  
getSubaddress(uint accountId, uint subaddressIdx): MoneroSubaddress

## Monero Wallet Interface 2/3

```
createSubaddress(uint accountId, string label=null): MoneroSubaddress
setAccountLabel(uint accountId, string label): void
setSubaddressLabel(uint accountId, string subaddressIdx, string label): void
getTx(string txId): MoneroTxWallet
getTxs(MoneroTxQuery query=null): MoneroTxWallet[]
getTransfers(uint accountId=null, uint subaddressIdx=null): MoneroTransfer[]
getOutputs(MoneroOutputQuery query=null): MoneroOutputWallet[]
getOutgoingTransfers(MoneroTransferQuery query=null): MoneroOutgoingTransfer[]
getIncomingTransfers(MoneroTransferQuery query=null): MoneroIncomingTransfer[]
getTransfers(MoneroTransferQuery query=null): void
exportOutputs(bool all=false): string
importOutputs(string outputsHex): uint
exportKeyImages(bool all=false): MoneroKeyImageExportResult
importKeyImages(MoneroKeyImage[] keyImages, uint offset=0): MoneroKeyImageImportResult
freezeOutput(String keyImage): void
thawOutput(String keyImage): void
isOutputFrozen(String keyImage): bool
getNewKeyImagesFromLastImport(): MoneroKeyImage[]
getDefaultFeePriority(): MoneroTxPriority
createTx(MoneroTxConfig config): MoneroTxWallet
createTxs(MoneroTxConfig config): MoneroTxWallet[]
sweepOutput(MoneroTxConfig config): MoneroTxWallet
sweepUnlocked(MoneroTxConfig config): MoneroTxWallet[]
sweepDust(bool relay=false): MoneroTxWallet[]
relayTx(string txMetadata): string
relayTx(MoneroTxWallet tx): string
relayTxs(string[] txMetadatas): string[]
relayTxs(MoneroTxWallet[] txs): string[]
describeTxSet(MoneroTxSet txSet): MoneroTxSet
signTxs(string unsignedTxHex): MoneroTxSet
submitTxs(string signedTxHex): string[]
signMessage(string message, MoneroMessageSignatureType type, int accountId, int subaddressIdx): string
verifyMessage(string message, string address, string signature): MoneroMessageSignatureResult
getTxKey(string txId): string
checkTxKey(string txId, string txKey, string address): MoneroCheckTx
getTxProof(String txId, string address, string message=null): string
checkTxProof(string txId, string address, string message, string signature): MoneroCheckTx
getSpendProof(string txId, string message=null): string
checkSpendProof(string txId, string message, string signature): bool
```

## Monero Wallet Interface 3/3

getReserveProofWallet(string message): string

getReserveProofAccount(uint accountId, ulong amount, string message): string

checkReserveProof(string address, string message, string signature): MoneroCheckReserve

setTxNotes(string[] txHashes, string[] notes): void

setTxNote(string txHash, string note): void

getTxNotes(string[] txHashes): string[]

getTxNote(string txHash): string

getAddressBookEntries(uint[] entryIndices=null): MoneroAddressBookEntry[]

addAddressBookEntry(string address, string description): uint

editAddressBookEntry(uint entryIdx, bool setAddress, string address, bool setDescription, string description): void

deleteAddressBookEntry(uint entryIdx): void

tagAccounts(string tag, uint[] accountIndices): void

untagAccounts(uint[] accountIndices): void

getAccountTags(): MoneroAccountTag[]

setAccountTagLabel(string tag, string label): void

getPaymentUri(MoneroTxConfig request): string

parsePaymentUri(string uri): MoneroSendRequest

getAttribute(string key): string

setAttribute(string key, string val): void

startMining(uint numThreads=null, bool backgroundMining=false, bool ignoreBattery=true): void

stopMining(): void

isMultisigImportNeeded(): bool

isMultisig(): bool

getMultisigInfo(): MoneroMultisigInfo

prepareMultisig(): string

makeMultisig(string[] multisigHexes, uint threshold, string password): string

exchangeMultisigKeys(string[] multisigHexes, string password): MoneroMultisigInitResult

exportMultisigHex(): string

importMultisigHex(string[] multisigHexes): uint

signMultisigTxHex(string multisigTxHex): MoneroMultisigSignResult

submitMultisig(string signedMultisigHex): string[]

moveTo(string path): void

changePassword(string oldPassword, string newPassword): void

save(): void

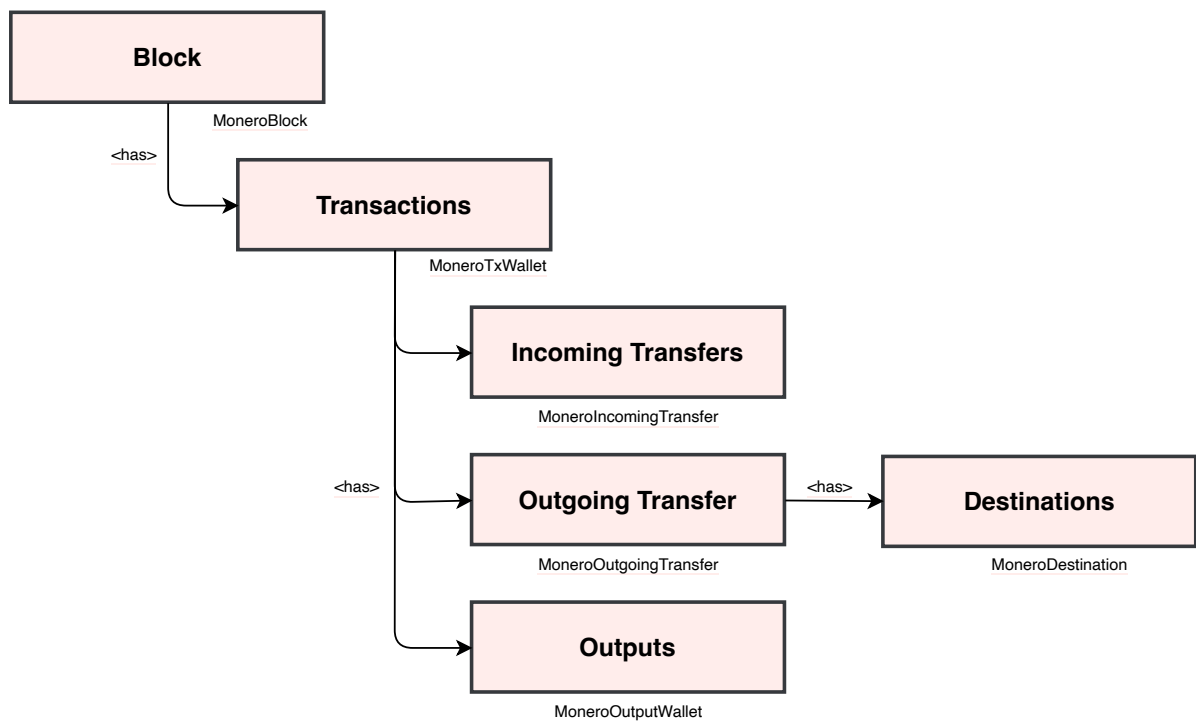
close(bool save=false): void

isClosed(): bool

getIntegratedAddress(string standardAddress=null, string paymentId=null): MoneroIntegratedAddress (deprecated)

decodeIntegratedAddress(string integratedAddress): MoneroIntegratedAddress (deprecated)

# Monero Wallet Types 1/4 - Transactions, Transfers, and Outputs



Transactions are part of a block and have incoming transfers and/or an outgoing transfer.

The outgoing transfer may have destination amounts and addresses which are saved locally with the wallet and are not recoverable from the blockchain.

Transactions may also have outputs received by the wallet when requested.

Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs

| MoneroTxConfig                    |
|-----------------------------------|
| address: string                   |
| amount: ulong                     |
| destinations: MoneroDestination[] |
| subtract_fee_from: int[]          |
| payment_id: string                |
| priority: MoneroTxPriority        |
| fee: ulong                        |
| account_index: uint               |
| subaddress_index: uint            |
| subaddress_indices: uint[]        |
| can_split: bool                   |
| relay: bool                       |
| note: string                      |
| recipient_name: string            |
| below_amount: ulong               |
| sweep_each_subaddress: bool       |
| key_image: string                 |

Configures creating transactions to send funds.

| MoneroOutputQuery extends MoneroOutputWallet |
|----------------------------------------------|
| subaddress_indices: uint[]                   |
| tx_query: MoneroTxQuery                      |
| min_amount: ulong                            |
| max_amount: ulong                            |

Configures a query to get wallet outputs.

| MoneroTxQuery extends MoneroTxWallet  |
|---------------------------------------|
| is_outgoing: bool                     |
| is_incoming: bool                     |
| hashes: string[]                      |
| has_payment_id: bool                  |
| payment_ids: string[]                 |
| height: ulong                         |
| min_height: ulong                     |
| max_height: ulong                     |
| include_outputs: bool                 |
| transfer_request: MoneroTransferQuery |
| output_request: MoneroOutputQuery     |

Configures a query to get wallet transactions.

| MoneroTransferQuery extends MoneroTransfer |
|--------------------------------------------|
| is_incoming: bool                          |
| address: string                            |
| addresses: string[]                        |
| subaddress_index: uint                     |
| subaddress_indices: uint[]                 |
| destinations: MoneroDestination[]          |
| has_destinations: bool                     |
| tx_query: MoneroTxQuery                    |

Configures a query to get wallet transfers.

## Monero Wallet Types 3/4

| MoneroAccount                    |
|----------------------------------|
| index: uint                      |
| primary_address: string          |
| balance: ulong                   |
| unlocked_balance: ulong          |
| subaddresses: MoneroSubaddress[] |
| tag: string                      |

| MoneroSubaddress            |
|-----------------------------|
| account_index: uint         |
| index: uint                 |
| address: string             |
| label: string               |
| balance: ulong              |
| unlocked_balance: ulong     |
| is_used: bool               |
| num_unspent_outputs: ulong  |
| num_blocks_to_unlock: ulong |

| MoneroDestination |
|-------------------|
| address: string   |
| amount: ulong     |

| MoneroOutputWallet<br>extends MoneroOutput |
|--------------------------------------------|
| account_index: uint                        |
| subaddress_index: uint                     |
| is_spent: bool                             |
| is_frozen: bool                            |

| MoneroTransfer      |
|---------------------|
| tx: MoneroTxWallet  |
| is_incoming: bool   |
| amount: ulong       |
| account_index: uint |

| MoneroTxSet             |
|-------------------------|
| txs: MoneroTxWallet[]   |
| multisig_tx_hex: string |
| unsigned_tx_hex: string |
| signed_tx_hex: string   |

| MoneroTxWallet extends MoneroTx      |
|--------------------------------------|
| tx_set: MoneroTxSet                  |
| incoming_amount: ulong               |
| outgoing_amount: ulong               |
| incoming_transfers: MoneroTransfer[] |
| outgoing_transfer: MoneroTransfer    |
| note: string                         |
| is_locked: bool                      |
| input_sum: ulong                     |
| output_sum: ulong                    |
| change_address: string               |
| change_amount: ulong                 |
| num_dummy_outputs: uint              |
| extra_hex: string                    |

Extends

Extends

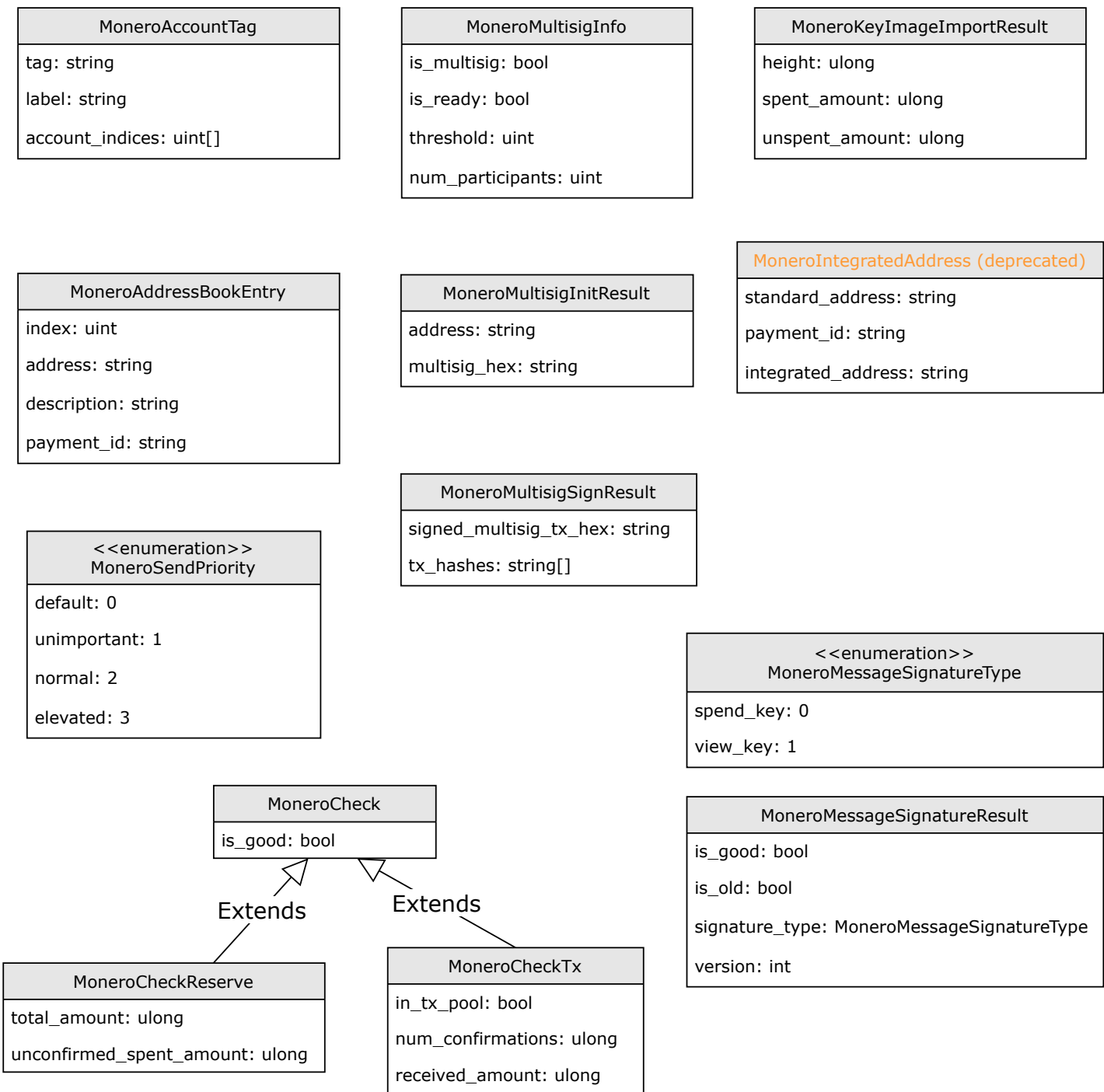
| MoneroIncomingTransfer             |
|------------------------------------|
| subaddress_index: uint             |
| address: string                    |
| num_suggested_confirmations: ulong |

| MoneroOutgoingTransfer            |
|-----------------------------------|
| subaddress_indices: uint[]        |
| addresses: string[]               |
| destinations: MoneroDestination[] |

| MoneroSyncResult          |
|---------------------------|
| num_blocks_fetched: ulong |
| received_money: bool      |

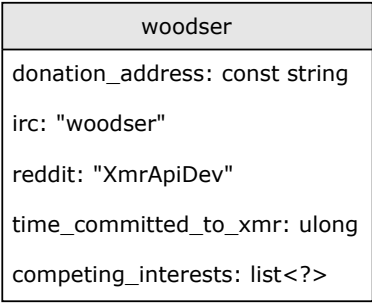
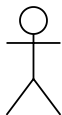
| MoneroWalletListener                                                                                       |
|------------------------------------------------------------------------------------------------------------|
| onSyncProgress(ulong height, ulong startHeight, ulong endHeight, double percentDone, string message): void |
| onNewBlock(ulong height): void                                                                             |
| onOutputReceived(MoneroOutputWallet output): void                                                          |
| onOutputSpent(MoneroOutputWallet output): void                                                             |

Monero Wallet Types 4/4





46FRlGKVqFNQnDiFkH7AuzbUBrGQwz2VdaXTDD4jcjRE8YkkoTYTmZ2Vohsz9gLSqkj5EM6ai9Q7sBoX4FPPYJdGKQQXPVz



Motivation tends to increase with support to donation address.